



CAI 1001C -2265-8449 Artificial Intelligence (AI)

Summer 2026

Modality: In Person | Credit Hours: 3.00

Homestead Campus | 06/22/2026-07/31/2026 MW 08:00 AM-11:20 AM

A survey of artificial intelligence (AI) where students explore different technologies utilizing concepts and skills widely accepted for AI and digital competency. Classification algorithms, supervised vs. unsupervised learning, data preparation, and training/using learning models for predictions are presented. CGS1060C and MAT1033 are recommended, but not required. (2 hr. lecture, 2 hr. lab)

Course Competencies and Learning Outcomes

Competency 1: Student will examine the field of AI and machine learning by:

1. Describing Digital Literacy, Practical Functional Digital Skill, Digital Collaboration, Curating Information, and Data Sources – Internet of Things
2. Describing the history of AI
3. Describing and using the AI project cycle
4. Identifying supervised vs. unsupervised learning, neural networks and discovering their application
5. Researching enterprise applications of AI and implementation readiness such as computing scalability, technical tradeoffs, and data pipelines

Competency 2: The student will demonstrate an understanding of ethical and legal considerations in AI by:

1. Defining key vocabulary terms related to computing ethics such as algorithmic privacy and algorithmic fairness
2. Identifying and explaining issues in computing ethics at the levels of system development, implementation, and administration

Competency 3: The student will explore data sets by:

1. Defining and describing data, data acquisition, data visualization, and data transformation.
2. Identifying data sources and their production
3. Using and manipulating data sets to train learning models

Competency 4: The student will explore machine- learning algorithms for classification by:

1. Describing classification algorithms including linear classifiers, support vector machines, quadratic classifiers, kernel estimation (k-nearest neighbor), decision trees, and neural networks
2. Reproducing linear classifiers, support vector machines, k-nearest neighbor, and neural net learning models using Python

Competency 5: The student will examine computer vision and facial recognition by:

1. Explaining computer vision and facial recognition, image segmentation, object and motion detection and object classification

Competency 6: The student will examine Natural Language Processing by:

1. Defining and explaining machine translation, sentiment analysis, application of deep learning to NLP, speech recognition, and synthesis
2. Reproducing Chatbots using Python

Competency 7: The student will examine Robotic Sensing and Manipulation by:

1. Defining and explaining robotics, sensing and manipulation, human-robot interaction, navigation and path planning, reinforcement learning, and autonomous vehicles technologies and impacts

Learning Outcomes:

1. Use quantitative analytical skills to evaluate and process numerical data
2. Solve problems using critical and creative thinking and scientific reasoning
3. Demonstrate knowledge of ethical thinking and its application to issues in society
4. Use computer and emerging technologies effectively

Required Textbook, Course Materials, or Technology

Required Materials

Computer

Internet Connection

Optional Course Materials

Additional materials will be given as the course progresses. No textbook is used in this course.

Assignment Schedule

Due Date	Assignment	Assignment Type	Points
	Class Question Board	Discussion	0

Course Grading Policy and Assessment Methods

Final grades are determined through a weighted average of quizzes, individual and group projects, laboratory assignments, class participation, and attendance. No late work is accepted. Your final grade in the course will be based on the following:

Grading policy and grade letters for the course

Category of Assignment	Grade Weights	Grade Letter	Percentage Range
Individual & Group Projects	25%	A	90 - 100%
Discussions	20%	B	80 - 89%
Quizzes	20%	C	70 - 79%
Project I, II, & III	10%	D	60 - 69%
Group Final Project	20%	F	0 - 60%
Attendance	5%		
Extra Credit	10%		

Incomplete grades are based solely on the instructor's discretion, require a signed agreement, and must receive the Department Chair's approval. Students must be passing the course to be eligible for entering *Agreements for Grade of Incomplete*.

MDC Grade Scale

Miami Dade College uses a standard grading scale based on letter grades.

MDC Grade Scale	
Grading Criteria	
Letter Grade	Percentage
A	90 - 100%
B	80 - 89%
C	70 - 79%
D	60 - 69%
F	0 - 59%

Attendance Policy

Participation and Attendance

Our course is given IN PERSON. This means we are meeting twice a week on the scheduled times for synchronous learning. You will require considerable time and effort outside of class time to finish the assigned homework activities. A suggested minimum of 6 hours per week is to be used in reviewing what you learned in class and applying it to your homework activities.

Students are expected to:

- **Maintain a regular weekly presence and respond to email, chats, surveys, polls, and any other in-class activities**
- **Participate regularly by submitting graded assignments on time**

Table describing attendance and participation policies.

Type	Description
Attendance	Taken every class. You are to be present on each session for at least 70% of the time to be counted as present.

Type	Description
Participation	Tracked by submissions of in-class exercises and by actively participating during class sessions.

If you stop attending this class (i.e., miss two consecutive class meetings or a total of 3 absences during the semester), or fail to submit assignments for two consecutive weeks, you may be dropped from the course without notice, and your record will show a grade of W (Withdrawal).

If you do not attend the first day of class, you will be automatically withdrawn from the course. This withdrawal will not appear on or affect your transcripts, nor will it leave a charge for the course. If you are withdrawn by me in the course, there will be no reinstatement.

Course Withdrawal

After registering, students may change their schedules during the drop/add period. The dates for this period are listed on the [Academic Calendar](#) that may be found as a link on the Miami Dade College homepage. At all times, your registration status is your responsibility. If you decide to withdraw, it is your responsibility to do so by this date.

Absences

Students are responsible for making up material, reading, and lab assignments covered during the missed class and must come prepared and up to date to class the next class meeting. The student must demonstrate satisfactory completion of any missed lab assignments to receive credit for them.

Miami Dade College Policies

[College Catalog](#)

Incomplete Grade Procedure

[Manual of Procedure: Granting and Removing of Incomplete Grades](#)

Student's Rights and Responsibilities

[Manual of Procedure: Student Rights and Responsibilities](#)

[MDC Student's Rights and Responsibilities Resource Website](#)

Student Support Services and Resources

- [Student Resources](#)
- [Tutoring Labs and Technology – Learning Resources](#)
- [Virtual Tutoring through Learning Resources](#) or [Free Online Tutoring through Brainfuse inside of your Canvas course](#)
- [Advisement](#)
- [Password and Login Technical Support](#)
- [Technical Support for MDC Live and MDC Online Courses](#)
- [SMART Plan](#)

ACCESS Disability Services

A Comprehensive Center for Exceptional Students' Services (ACCESS) - Services for Students With Disabilities

Miami Dade College provides support services and accommodations to assist students with documented disabilities achieve their education goals. You may learn about these services and eligibility by contacting your [campus's ACCESS](#) office. **No retroactive accommodations will be provided.**

For more information, please visit [Miami Dade College Accessibility Statement, procedure 4055](#).

Additional Support Services and Resources

- [Public Safety - Services](#)
- [Hurricane and Other Natural Disasters](#): In the event of a hurricane or other disaster, the class follows the schedule established by the College for campus-based courses. Please visit the [MDC website](#) or call the MDC Hotline (305-237-7500) for situation updates.